

Fish and flow management in the Murray–Darling Basin: Directions for research

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Summary Effective natural resource management requires knowledge exchange between researchers and managers to support evidence-based decision-making. To achieve this, there is a need to align research with management and policy needs. This project aimed to identify the flow-related ecological knowledge needs for freshwater fish to better inform environmental water management in the Murray–Darling Basin, south-eastern Australia. Our major objective was to provide an up-to-date assessment of scientific research and integrate this with the knowledge requirements of relevant managers to guide future research. We reviewed the contemporary scientific literature and engaged managers specifically responsible for delivering flows for fish outcomes via a questionnaire and workshop. Research on fishes of the MDB has generally evolved from single locations and/or times to larger spatio-temporal scales, including multiple sites, rivers and catchments. There has also been a trend from single life stage studies to incorporation of multiple life stages and population processes. There remain, however, significant deficiencies in knowledge for most native species, many of which are threatened. Four agreed key knowledge gaps were derived from the literature review and managers' suggestions: (i) population dynamics, (ii) movement, dispersal and connectivity, (iii) survival and recruitment to adults and (iv) recruitment drivers. To inform policy and management, managers desired timely advice, based on robust research and monitoring. Fish species of most relevance to managers were those highly regarded by community stakeholders and whose life histories and population dynamics are potentially influenced by flow. Populations of these mostly large-bodied, angling species (e.g. Murray Cod, Golden Perch and Silver Perch) have declined, often due to river regulation and, in conjunction with managers' priorities, are relevant candidates for research to support the management of flow to rehabilitate fish populations in the MDB.

Key words: collaborative workshop, environmental water, fish and flow ecology, literature review, policy, natural resource management.

Introduction

Effective natural resource management requires timely and appropriate knowledge exchange between researchers and managers to support evidence-based decision-making (Cvitanovic *et al.* 2015). Limited resources and time frames, and a lack of access to pertinent information, however, mean that the ecological knowledge used in management is often less than ideal (Pullin *et al.* 2003). Under these constraints, there is a need to improve mechanisms to ensure that the best available science is relevant and accessible to policy and management (Rose *et al.* 2018). Fundamental to this is the establishment of relationships between researchers and managers that foster the acquisition and dissemination of science (Safford *et al.* 2017).

Worldwide, freshwater fish are subject to a multitude of anthropogenic stressors,

including river regulation (Dudgeon *et al.* 2006; Closs *et al.* 2015). The Murray–Darling Basin (MDB) in south-eastern Australia is one of the most regulated riverine systems in the world (Nilsson *et al.* 2005). The life cycles of riverine fishes are inextricably linked to flow, and the alteration of flow regimes in the MDB poses a key threat to populations (Koehn & Lintermans 2012). MDB fish populations have suffered substantial declines and, overall, are now estimated to be at about 10% of pre-European levels (MDBC 2004; Koehn *et al.* 2014a), with many species considered threatened (Lintermans 2007). Fishes of the MDB have important ecological, biodiversity, scientific, cultural and recreational values. For example, recreational fishing is a significant pastime, with 19% of Australians participating annually (Henry & Lyle 2003), providing substantial social and economic benefits to rural

communities (\$1.3 billion per annum; Ernst & Young 2011). The poor state of native fish populations in the MDB provides an important context for water management, including a need to restore fish populations, rather than maintain their current depauperate state (Koehn & Lintermans 2012). Delivery of environmental water allocations is now a well-recognised management action in the rehabilitation of rivers and their fishes (Arthington & Pusey 2003; Arthington 2012). In the MDB, the Basin Plan (MDBA 2010, 2011) establishes fishes as important ecological assets, identifies environmental flows as vital for their conservation, and reallocates water for environmental rehabilitation (Koehn *et al.* 2014b).

The management of environmental water for fish-related outcomes has developed rapidly in the 21st century, especially in the MDB, and this has been

accompanied by a need for knowledge and understanding of the flow-related ecology of fishes (Koehn *et al.* 2014c). Whilst there are many examples of new research (e.g. Beesley *et al.* 2011; King *et al.* 2016; Tonkin *et al.* 2017a,b), knowledge syntheses (e.g. Bice *et al.* 2014; Koehn *et al.* 2014b,c; NSW DPI 2014, 2015a,b, Ellis *et al.* 2015) and directions for management (e.g. King *et al.* 2010; Koehn *et al.* 2014c; Mallen-Cooper & Zampatti 2015), there remains a need for additional research to fill knowledge gaps and refine management. The need for knowledge to be incorporated into management objectives and decisions is a priority, yet the effective transfer of such knowledge is also a current constraint (Koehn *et al.* 2014c; Horne *et al.* 2017). Given the commitment to the Basin Plan and the increased availability of environmental water (MDBA 2010, 2011), there will be increased public scrutiny of the use of water for ecological outcomes and an expectation of demonstrable environmental benefits.

Whilst there have been advances in scientific knowledge regarding the ecology of Australian freshwater fishes (e.g. Humphries & Walker 2013), an understanding of how to best allocate water to achieve native fish outcomes is still emerging (Koehn *et al.* 2014b). In particular, greater knowledge is required relating flow (hydrology and hydraulics, *sensu* Mallen-Cooper & Zampatti 2018) to fish life histories and population dynamics, and how flow management can be used to maximise ecological benefits for fish. For successful environmental flow management in the MDB, it is imperative that research that aims to develop knowledge to underpin management is conceived collaboratively between researchers and managers.

In this study, we aimed to:

- 1 Review contemporary peer-reviewed knowledge regarding the flow-related ecology of MDB fishes to identify scientific research trends.
- 2 Use a collaborative approach to gain an understanding of the knowledge needs of managers responsible for delivering flows for fish outcomes across the MDB.
- 3 Assess the level of knowledge and identify knowledge gaps concerning the flow-related ecology of MDB fishes relevant to the fish objectives of the Basin Plan.
- 4 Identify future research directions that will directly support water management to meet the fish objectives of the Basin Plan and associated watering strategies.

Methods

Review of ecological knowledge

To review ecological knowledge and research trends regarding the flow-related ecology of freshwater and estuarine native fishes in the MDB, we integrated information from (i) recent published reviews of the literature incorporating peer-reviewed and grey literature and (ii) a contemporary review of peer-reviewed literature using the ISI-Web of Science. Several comprehensive reviews of flow-related fish ecology have been undertaken by the Murray–Darling Basin Authority (MDBA) and State agencies (e.g. King *et al.* 2009; Bice *et al.* 2014; Koehn *et al.* 2014c; NSW DPI 2014, 2015a,b, Ellis *et al.* 2015), and these were reviewed for relevant information and used to better target literature review search terms. The peer-reviewed literature search was then undertaken to examine research trends from 1900 to present, specifically concerning the flow-related ecology of fish in the MDB. We searched the ISI–Web of Science on 13/04/2016—just prior to

the management workshop (see below), using the following terms: *Australia* and *fish* and *flow*. Given the large number of articles using only these three search terms (745), we then used the same three terms again each in combination with *population*, *spawning*, *growth*, *movement*, *recruitment*, *condition* and *habitat*. For each of these final search combinations, we read the titles, keywords and abstracts to develop a final list (Table 1). Papers were excluded if: (i) the study was not directly investigating flow-related fish ecology; (ii) the study was not on native fish species; (iii) the study was not in the MDB (unless it was in an adjacent catchment with a shared MDB species); and (iv) the manuscript was a data modelling paper rather than containing field generated data. We also captured information relating to the scale of study, macrohabitat type (e.g. river, wetland and estuary), location, species and life stages investigated. The search only included ISI registered journals and fully published papers (not early online), and by only interrogating the abstract, titles and keywords, we recognise that some studies may have been missed (especially for multispecies papers).

The results of the ISI database search were combined with prior literature reviews to determine the knowledge available in relation to each species' life-history processes, population dynamics and recruitment drivers. From this, we briefly assessed the relative amount of knowledge regarding the flow-related biology and ecology of key fish species in the MDB. Species were selected due to their presence in catchments where flow is

Table 1. Search terms, total and relevant number of papers from the ISI literature search (Note: individual papers can address more than one search term combination—total relevant papers = 68)

Search terms	Total number of papers	Number of relevant papers
Australia & fish & flow	745	
Australia & fish & flow & population	319	13
Australia & fish & flow & spawning	106	19
Australia & fish & flow & growth	84	13
Australia & fish & flow & movement	101	19
Australia & fish & flow & recruitment	139	29
Australia & fish & flow & condition	195	13
Australia & fish & flow & habitat	323	14

managed for fish outcomes and their relevance as target species for managers. The status of existing knowledge for each life stage, of each species, was ranked from 1 to 10 (1 reflecting limited and 10, extensive knowledge), by the three authors (each independently and then by consensus where there was disagreement).

Engagement with fish and flow managers

We engaged individuals and organisations specifically responsible for managing environmental water and flows to achieve fish outcomes. The engagement entailed three components: (i) a preworkshop questionnaire; (ii) a workshop with invites to fish researchers (ecologists), fish and fisheries managers (managers of conservation and fisheries outcomes—hereafter referred to as fish managers) and flow managers; and (iii) a postworkshop review of outcomes and priorities. To structure the workshop and to allow for opinions unbiased by workshop discussions, we provided a questionnaire to invitees one month prior to the workshop. We also provided participants with the option to complete the questionnaire even if workshop attendance was not possible. The questionnaire (see Appendix S1) sought to elucidate the perspectives of fish and flow managers in relation to flow-related fish ecology or fish outcomes. In particular, this included how water was managed for fish outcomes, including relevant spatial and temporal scales; what knowledge/information was needed by managers; past or future changes in management and knowledge needs; priority fish species; risks and constraints to management; and what managers want from fish researchers.

The workshop was conducted on 5 May 2016 and included an update on current knowledge pertaining to the flow-related ecology of fishes in the MDB, a report on the literature review, and case study presentations from fish and flow managers from each Murray–Darling Basin State jurisdiction plus the Murray–Darling Basin Authority and the Commonwealth Environmental Water Office. Responses to the questionnaire were tabled and discussed, and a summary of research

priorities workshopped to develop a consensus on priority knowledge gaps and management needs. These agreed knowledge gaps were then collated and sent to participants after the workshop for further consideration and agreement, to ensure they were an accurate reflection of the workshop outcomes. The outcomes of both the literature review and management needs workshop were compared to examine similarities and discrepancies in research and management needs for the effective flow-related management of MDB fishes.

Results

Review of ecological knowledge

The literature search revealed 68 papers that met the search criteria (Table 1, Appendix S2) primarily concerning the large-bodied species, Murray Cod (*Maccullochella peelii*), Golden Perch (*Macquaria ambigua*) and Black Bream (*Acanthopagrus butcheri*) (a marine–estuarine species, with most studies not from the MDB). Studies were largely limited to a few regions, particularly the lowland Murray River (downstream of the Yarrowonga Weir), with the northern MDB poorly represented (12 of 68 papers) compared to the southern MDB. Most studies regarding the flow-related ecology of fishes in the MDB have occurred since 2008. The general trends through time are summarised in Figure 1, where the 68 papers are categorised into (i) the life stages that were investigated (top panel), (ii) the drivers that are likely to influence population demographics of native fish and are themselves affected by flow (middle panel), and (iii) those studies that directly investigated population demographics (bottom panel). In relation to life stages, there were more studies on juveniles and adults compared to eggs and/or larvae, with a consistent output of papers on juvenile and adult studies from 2006 onwards.

Prior to 2008, most flow-related fish ecology studies examined flow variables that could influence the likelihood of spawning and recruitment. Since 2008,

studies have examined the relationships between flow and a range of other parameters including growth, body condition and movement (Fig. 1). Of these recent papers, however, several were estuarine studies, with a general paucity of research in riverine environments. Few papers dealt explicitly with environmental flows and the management of native fishes. The spatial scales of studies were usually limited to single sites/river reaches, and temporal scales were similarly limited to short timescales (e.g. one or two sampling occasions or single-year studies).

Flow-related ecological knowledge, across all life stages, is greatest for large-bodied species, particularly Murray Cod, Trout Cod (*Maccullochella macquariensis*) and Golden Perch, and least for small-bodied fishes and juvenile life stages of most species (Table 2). Of the large-bodied fishes, knowledge regarding eggs and larvae has been derived from studies both in the hatcheries and in the wild, and is greatest for Murray Cod and Trout Cod.

Engagement with fish and flow managers

The fish and flow manager's workshop was attended by 19 participants (six fish researchers, two fish managers, eight flow managers and three project managers), whilst 13 flow managers provided preworkshop questionnaire responses (Appendix S1). A resounding objective from managers was the need for *more fish*. Specifically, this incorporated increased fish abundance across species, achieved by improving survivorship through to adults, and increased spatial distribution. Managers were satisfied to use interim measures such as the presence of early life stages to indicate progress towards longer-term goals but needed to be able to illustrate to their stakeholders (regional public, irrigators and anglers) the overall benefits of their actions to fish populations.

Flow management knowledge needs

Water management for fish outcomes in the MDB occurs at a range of spatial (site to whole-of-river) and temporal scales (from annual watering priorities to

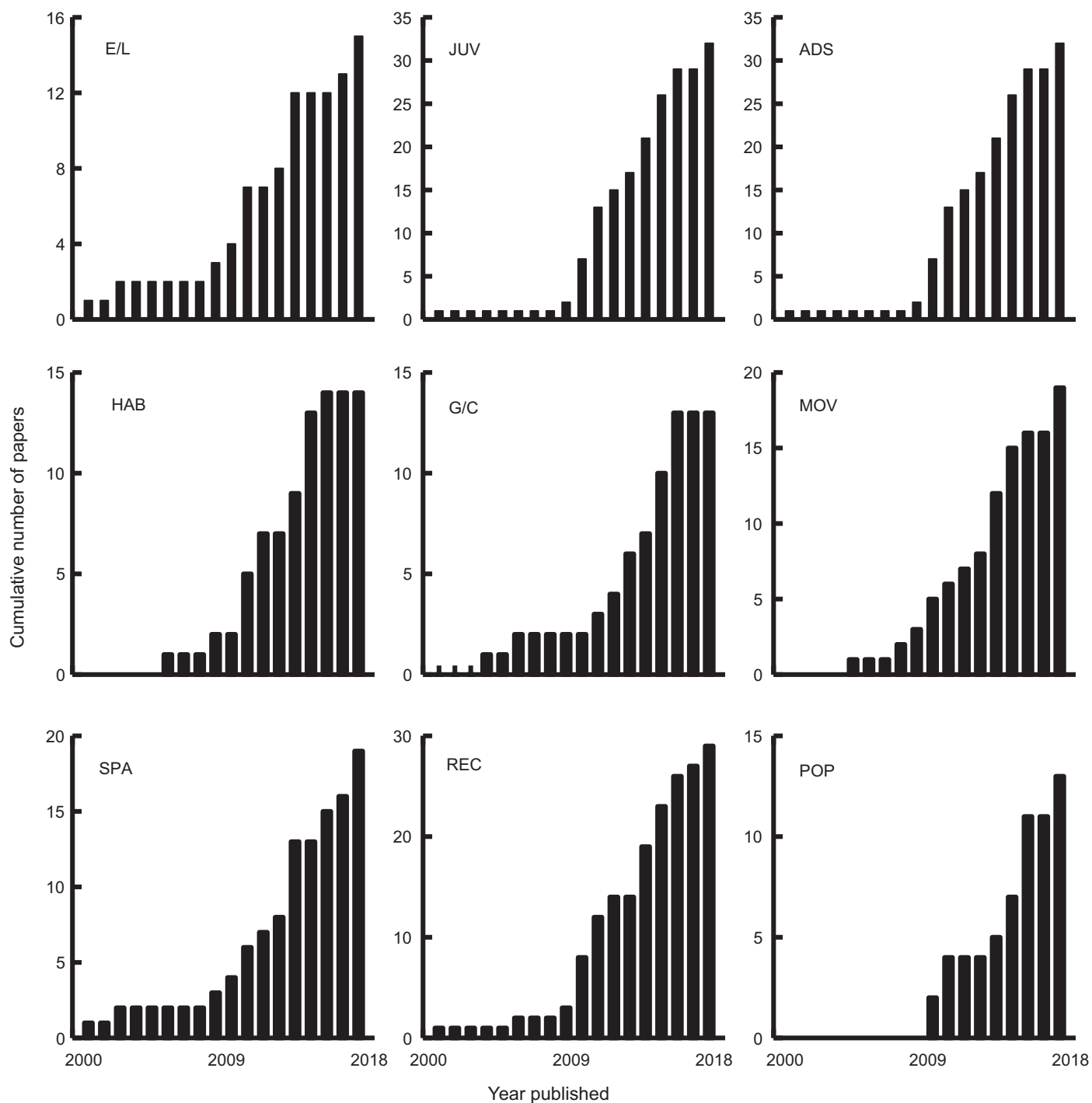


Figure 1. Cumulative number of published peer-reviewed papers from the Web of Science (2000–2018) regarding the flow-related ecology of fish relevant to the Murray–Darling Basin. Plots arranged for papers investigating three life stages: top panel—eggs or larvae (E/L), juveniles (JUV) and adults (ADS); middle panel—three aspects important for recruitment outcomes: habitat (HAB), growth or condition (G/C), movement (MOV); bottom panel—three key aspects of fish reproduction outcomes: spawning (SPA), recruitment (REC) and population (POP).

decadal water plans). The information used to inform management decisions includes data from monitoring, advice from fisheries managers and fish researchers, and local knowledge. Quantitative

population or conceptual ecological models may also be used. The typical hydrological scenarios utilised to achieve fish outcomes include the delivery or conservation of flow to inundate benches,

maintain higher baseflows, promote spawning and recruitment of large-bodied native species (mostly in late spring and summer) and provide movement cues to assist dispersal. Management needs have

Table 2. Relative ecological knowledge for key species life stages (from peer review and grey literature). Scores range from 1 to 10, with 1 indicating limited knowledge and 10 considerable knowledge

Species	Spawning	Eggs	Larvae	Juveniles	Adults
Murray Cod	9	9	8	6	9
Trout Cod	9	9	8	6	9
Freshwater Catfish	7	4	4	3	6
Golden Perch	6	7	5	4	8
Macquarie Perch	6	7	5	4	9
Silver Perch	6	7	5	4	7
Olive Perchlet	3	4	3	3	5
Purple-Spotted Gudgeon	4	4	3	3	5
Southern Pygmy Perch	4	4	3	3	5
Yarra Pygmy Perch	3	4	3	3	5
Murray Hardyhead	4	4	3	3	6

changed rapidly since the implementation of the Basin Plan and now incorporate multiple life stages and population outcomes rather than individual stage processes (e.g. spawning); increased monitoring to determine causal relationships; and the inclusion of connectivity, movements and spatial scales as important ecological components for populations. Future changes expected by managers include more monitoring, use of new knowledge, refinements to objectives and management processes, increased landscape and system-scale management (cross-jurisdictional), better understanding and the ability to make management trade-offs (such as targeting different species or life stages in different years) and further application of adaptive management (*sensu* Walters & Holling 1990).

There were synergies between the knowledge gaps identified from the literature review and manager's priority knowledge needs. Both fish researchers and managers highlighted the importance of understanding the processes that influence the dynamics of populations (i.e. recruitment, mortality and movement) and the spatio-temporal scales over which these processes occur. Four key ecological knowledge gaps were agreed to (i) population dynamics (incorporating all life stages); (ii) movement, dispersal and connectivity (incorporating spatial and temporal perspectives); (iii) recruitment to adults, through survival and growth; and (iv) mechanisms, causal links, thresholds and drivers (Table 3). Integrated studies that consider the influence of flow on key

population processes were considered the highest priority for research concerning the flow-related ecology of fish and management of environmental water in the MDB. The range of secondary priorities raised by managers (Table 3) related to the type of relationships, ecological knowledge and assistance that they are seeking from fish researchers. This led to a range of discussions at the workshop recognising the following topics: collaborative relationships; the need to build on other research and applied management projects; a predictive capacity; robust causal relationships to increase confidence in environmental water delivery; availability and the use of process-based population models; understanding the scale at which populations operate, demographic processes, connectivity; recruitment to adults being the key measure of success; incorporation of food webs, primary productivity, and food resources into management decisions; flow-related thresholds, less than optimum durations, partial events and flow sequences; Carp (*Cyprinus carpio*) (an alien species and hence not include in our literature review); trade-off processes; recovery times and recolonisation rates (e.g. from drought or blackwater); flow thresholds and sites for refugia, intermittent rivers; and that the northern MDB requires a greater research focus.

What managers expect from fish researchers

Fish and flow managers expect the following from fish researchers: the provision of robust scientific knowledge based on

research and monitoring; timely, relevant advice presented in simple terms; be prepared to 'give things a go'; a proactive, collaborative approach to determining research priorities and water proposals; advice regarding hydrographs (see below); and information on the potential impacts of Carp, including the potential release of the Koi herpes virus (McColl *et al.* 2014). In general, managers want cost-effective science to be undertaken by researchers that have the appropriate skills, capability and capacity, with research programs preferably developed in conjunction with the managers who need to utilise the outcomes. New knowledge needs to be transferred to managers, and a strategy for doing this should be included within research projects. Scientifically validated, causal relationships for fish responses to flow was a key request, but importantly, managers stressed that there was as an immediate (not aspirational) need for information that can be applied to current management questions.

Hydrographs that incorporate specific, ecologically relevant flow components (e.g. Yarnell *et al.* 2015) are being explored by fish and flow managers in the southern MDB to help plan water delivery to achieve fish objectives (e.g. see Stuart & Sharpe 2017). These hydrographs are generally constructed on an annual basis to reproduce flow components that may influence the life cycle of selected species. Managers are seeking relevant science and advice to support the development of these hydrographs.

Risks and constraints to management

Managers indicated a range of actual or perceived risks that influenced their management of environmental water. These included uncertainties in the knowledge used in decision-making, increased Carp abundance, poor water quality and the potential for fish kills, and not achieving expected outcomes. There was also a range of constraints when delivering environmental water: water availability, time, competing demands for water (e.g. irrigation needs); operational considerations; barriers to movement (e.g. adequate

Table 3. A summary of priority knowledge gaps identified in the fish and flow manager's workshop and agreed upon by the participants

Ecological knowledge gaps (determined by the ISI review and by fish researchers)	
Highest priorities	Secondary priorities
1 Population dynamics (incorporating all life stages)	• Fish condition (and effects on survival and recruitment)
2 Spatial and temporal scales and population processes	• Fecundity
3 Survival, growth rates	
4 Recruitment drivers (food, etc.)	
Management knowledge gaps (provided by managers)	
Highest priorities	Secondary priorities
1 Population dynamics	• Trade-off processes
2 Recruitment to adults	• Species-specific responses to flows
3 Movement, dispersal and connectivity	• Life stage-specific responses to flows
4 Mechanisms/causal links and thresholds (scale of variability; what are the drivers)	• Scale—landscape/system—site
	• Refugia flow thresholds, maintain or not, top up or not
	• Recovery time (drought/blackwater)—recolonisation, barriers

flows for fishways); and public concern regarding risks, especially relating to third-party impacts such as flooding.

Manager's priority species

Priority fish species for managers could be broadly delineated into large and small-bodied fishes. The large-bodied fishes were dominated by publicly popular native species, which in priority order, were as follows: Murray Cod, Golden Perch, Trout Cod, Silver Perch (*Bidyanus bidyanus*), Macquarie Perch (*Macquaria australasica*) and Freshwater Catfish (*Tandanus tandanus*). Small-bodied fish, in priority order, were as follows: Southern Pygmy Perch (*Nannoperca australis*), Southern Purple-Spotted Gudgeon (*Mogurnda adspersa*), Olive Perchlet (*Ambassis agassizii*), Murray Hardyhead (*Craterocephalus fluviatilis*) and Yarra Pygmy Perch (*Nannoperca obscura*). Carp were also considered important, especially in relation to potential recruitment and population growth associated with environmental flows, either real or perceived, with a particular interest in

the potential impacts of the Koi herpes virus on Carp populations, should this be released (McColl *et al.* 2014). Whilst managers often prioritise large-bodied native fishes of public relevance, there is also consideration of threatened small-bodied fishes and whole fish communities, which may be addressed through consideration of umbrella or keystone species. Estuarine and diadromous species form specific priorities in the lower Murray River in South Australia, and their inclusion in broader environmental water planning was supported by representatives from all MDB jurisdictions.

Discussion

In Australia's MDB, increased environmental water allocations through the Basin Plan (MDBA 2010, 2011) and other programs are being used to rehabilitate degraded native fish populations (Koehn *et al.* 2014b; Stoffels *et al.* 2018). For nature conservation to be effective, natural resource managers require access to pertinent science. The generation and uptake

of scientific knowledge is most successful when supported by robust and mutually beneficial partnerships between scientists and managers (Safford *et al.* 2017). In this study, we reviewed the scientific literature and sought direct input from fish and flow managers in the MDB, to identify priority knowledge gaps concerning the flow-related ecology of native fishes. These agreed knowledge gaps provide a basis to guide research to better inform the management of flow in the MDB for fish outcomes.

The state of knowledge

Research concerning the flow-related ecology of native fishes in the MDB has progressed substantially since 2008, prior to which there were few peer-reviewed journal articles relating flow to fish population outcomes. This may reflect that the first use of environmental water with explicit objectives for fish outcomes only occurred in 2005 (Koehn *et al.* 2014b). Research has generally evolved from single life stages (e.g. the influence of flow on spawning) and sites/ivers, to now include recruitment and the dynamics of populations across broad spatio-temporal scales. As part of this, the application of new technology, including otolith chemical analysis and population genomics, is allowing for a greater understanding of population processes and the spatial extent of populations (e.g. Zampatti *et al.* 2015; Beheregaray *et al.* 2017; Crook *et al.* 2017).

A key knowledge gap for most native fishes in the MDB is the influence of flow on survival rates between life stages contributing to population dynamics. Collective increases in survival through life stages ultimately lead to the increases in adult fish abundance and population growth—a key objective of environmental water management. If survival and recruitment can be related to flow, this can then be used to predict the trajectories of populations and illustrate the benefit of environmental flows (Shenton *et al.* 2012; Todd *et al.* 2017). In studies of the flow-related ecology of fishes, a range of definitions of recruitment is often used (e.g. to the post-larval life stages, to age 0+, into a fishery). To avoid confusion, the stage

to which recruitment is being referred should be clearly stated in flow-recruitment studies. Whilst the presence of specific life-history stages may be useful interim measures for outcomes from environmental water management, a combined range of factors influences the dynamics of populations and recruitment to adults. For example, links between flow, habitat and fish condition are known to influence recruitment, but have received little attention (see Tonkin *et al.* 2011, 2017a). Whilst these potential drivers of recruitment are fundamental for our understanding of population dynamics, they persist as knowledge needs for many species.

The movement of fish is another process central to the structuring of populations (Lucas & Baras 2001). Although there is now recognition of the need to incorporate movements of all life stages into the understanding of populations, there are few studies that have quantified the impacts of flows upon them. Spatial scales of movements vary between life stages and species. Flow-influenced movements include the passive drift of eggs or larvae, and the active movement of juveniles and adults (see Koehn & Crook 2013). As movement is a fundamental population processes, quantification of rates of movement and recolonisation are needed.

In general, studies of the flow-related ecology of fishes in the MDB have been undertaken at relatively few key geographic locations and the spatial transferability of results remains an issue to be resolved. The northern MDB is the most deficient in knowledge (see also NSW DPI 2014) and transferring understanding from studies in the southern MDB may be inappropriate due to aspects such as climate (rainfall seasonality) and flow intermittency (Balcombe *et al.* 2014).

Research priorities

By necessity, fish and flow managers tend to focus on outcomes for large-bodied native fishes of cultural and recreational relevance (e.g. Golden Perch and Murray Cod). In addition, they may also consider threatened species (e.g. Silver Perch), umbrella or keystone species (e.g. Murray

Cod) or the whole of fish community. If managers are seeking to use flow to achieve population-level outcomes for native fish, research to inform management should focus on those species whose key life-history processes (e.g. recruitment and movement) have been associated with flow. In this respect, Golden and Silver Perch, as widespread and wide-ranging species, provide ideal candidates. The rehabilitation of Silver Perch populations is a high priority (Koehn *et al.* 2014c; Tonkin *et al.* 2017c), and the recreational angling appeal of Golden Perch mean that both species form a considerable focus in contemporary environmental flow management in the MDB. Diadromous species such as Pouched Lamprey (*Geotria australis*) and Congolli (Tupong) (*Pseudaphritis urvillii*) have recruitment and movements associated with flow and connectivity and have become a focus in recent years (MDBA 2017).

The flow-related ecology of small-bodied fishes in the MDB remains a significant knowledge gap, potentially due to a lack of recognition and difficulties in conducting research on these species (e.g. low numbers, fragmented populations). Given that many of these species are threatened and have formed the focus for environmental water management (e.g. Murray Hardyhead and Yarra Pygmy Perch; Hammer *et al.* 2013), these species should be considered potential research candidates as a component of a holistic approach to the rehabilitation of native fish populations in the MDB.

There was a significant convergence between contemporary ecological knowledge, including deficiencies, and the knowledge requirements of water managers. The four key ecological knowledge gaps identified and agreed to (i) population dynamics; (ii) movement, dispersal and connectivity; (iii) recruitment; and (iv) mechanisms, causal links, thresholds and recruitment drivers, represent some of the broad factors that influence population growth and can guide the evolution of research to improve the management of fish populations using environmental water. Whilst research is fundamental to answering ecological questions and supporting management actions, it is imperative that research outcomes can be readily

interpreted and used to assist managers in their decision-making. Furthermore, as investment in environmental water increases, so will public scrutiny; therefore, it is critical that research is aligned with the needs of fish and flow managers to support decisions and demonstrate outcomes.

The future

To assist in decision-making, fish and flow managers expressed clear expectations of researchers. Specifically, they want scientifically validated causal relationships for fish responses to flow that can ultimately be used to inform the management of water (environmental allocations, natural flows or irrigation releases) for positive fish outcomes. To achieve this, managers want to foster collaborative relationships with researchers, to ensure research is relevant to management needs and is also aligned with knowledge deficiencies. Furthermore, managers want the outcomes of research articulated in an easily understandable 'plain English' form, something that published scientific papers alone may not achieve. Being aware of manager's needs will focus research and facilitate uptake, thus improving the ecological outcomes of water management.

Managers want fish researchers to have an 'applied' approach to research and be prepared to 'give things a go'. Both fish researchers and managers should embrace 'Adaptive Management' and see restoration activities such as water and flow management as scientific 'experiments' from which we can learn (Gellie *et al.* 2018). This enforces the need to not only provide water to achieve positive population outcomes for fish, but also to plan scientifically and monitor rigorously. A proactive, collaborative approach to determining research priorities and collegiate development of watering plans is key to advancing the field of environmental water management. Indeed, knowledge generated by research is likely to be more useful when there is significant interaction and knowledge sharing between fish researchers and research recipients (including stakeholders), particularly over the longer term (Chapman *et al.* 2017).

Other key stressors that may limit the effectiveness of flow management (e.g.

connectivity) also need to be considered, and indeed, a holistic approach to fish rehabilitation that includes flows and complementary measures, such as habitat restoration and improved connectivity, has long been advocated (Koehn & Lintermans 2012). The collation of contemporary ecological knowledge allows for the development of conceptual models for fish, which can be integrated with population models to enable managers to explicitly explore the likely outcomes and risks of their actions within an adaptive management framework (Koehn *et al.* 2018). Such models, however, rely on the results from fundamental research and monitoring of interventions.

Conclusion

Through consensus between ecologists and managers, this project has provided a focus on priority research for understanding the flow-related ecology and population dynamics of fish in the MDB. This study assists flow management for fish by outlining research deficiencies and priorities that are aligned with the needs of managers who are seeking flow-related fish outcomes. Whilst knowledge concerning the flow-related ecology of some fish in the MDB has progressed, in particular for large-bodied fishes, there remain considerable deficiencies. Factors such as the relative rates of survival and growth for all life stages of most species are essential to understand how to improve adult populations—the primary objective of managers. Research undertaken in collaboration with managers will increase utility and uptake, as will the communication of results in a manner digestible to a non-scientific audience. Developing this knowledge and implementing it in an adaptive management framework are fundamental to achieving the fish objectives of the Basin Plan and its associated watering strategies.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Appendix S1 The fish and flows manager's pre-workshop questionnaire (questions in bold) and a brief summary of responses.

Appendix S2 A list of the relevant fish and flow-related ecology related references extracted from ISI literature Review.